

Stylianos Loukas Vasileiou

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RESEARCH INTERESTS

Human-Aware AI: Explainable & Collaborative Decision-Making, User Modeling

Knowledge Representation & Reasoning: Applied Reasoning, Argumentation, Planning, Belief Revision

Other: Neuro-symbolic AI

EDUCATION

PH.D. IN COMPUTER SCIENCE
Washington University in St. Louis

FL' 2018 – SP' 2025
St. Louis, USA

M.S. IN ARTIFICIAL INTELLIGENCE
University of Southampton

FL' 2016 – FL' 2017
Southampton, UK

M.S. IN APPLIED MATHEMATICS
University of Glasgow

FL' 2015 – FL' 2016
Glasgow, UK

B.S. IN STATISTICS AND ACTUARIAL-FINANCIAL MATHEMATICS
University of the Aegean

FL' 2010 – FL' 2014
Samos, GR

PROFESSIONAL EXPERIENCE

NEW MEXICO STATE UNIVERSITY
Assistant Professor, Department of Computer Science

2025 –
Las Cruces, USA

PUBLICATIONS

Journal Articles

- [1] **Stylianos Loukas Vasileiou**, William Yeoh, Alessandro Previti, and Tran Cao Son. “On Generating Monolithic and Model Reconciling Explanations in Probabilistic Scenarios”. In: *Journal of Artificial Intelligence Research (JAIR)* 84 (2025).
- [2] **Stylianos Loukas Vasileiou**, William Yeoh, Tran Cao Son, Ashwin Kumar, Michael Cashmore, and Dianele Magazzeni. “A logic-based explanation generation framework for classical and hybrid planning problems”. In: *Journal of Artificial Intelligence Research (JAIR)* 73 (2022).

Conference Papers

- [1] **Stylianos Loukas Vasileiou**, Antonio Rago, Francesca Toni, and William Yeoh. “Argumentative Human-AI Decision-Making: Toward AI Agents That Reason With Us, Not For Us”. In: *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) BlueSky Track*. 2026.
- [2] Silvia Tulli, **Stylianos Loukas Vasileiou**, Mohamed Chetouani, and Sarath Sreedharan. “Inferring Implicit Goals Across Differing Task Models”. In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2026.
- [3] Antonio Rago, **Stylianos Loukas Vasileiou**, Francesca Toni, William Yeoh, and Tran Cao Son. “A Methodology for Incompleteness-Tolerant and Modular Gradual Semantics for Argumentative Statement Graphs”. In: *International Conference on Principles of Knowledge Representation and Reasoning (KR)*. 2025.
- [4] **Stylianos Loukas Vasileiou** and William Yeoh. “TRACE-cs: A Hybrid Logic-LLM System for Explainable Course Scheduling”. In: *International Conference on Principles of Knowledge Representation and Reasoning (KR)*. 2025.

- [5] **Stylianou Loukas Vasileiou** and William Yeoh. “TRACE-cs: A Synergistic Approach to Explainable Course Scheduling Using LLMs and Logic”. In: *AAAI Conference on Artificial Intelligence (AAAI) (Demo Track)*. 2025.
- [6] Yinxu Tang, **Stylianou Loukas Vasileiou**, Derkinderen Vincent, and William Yeoh. “Model Reconciliation via Cost-Optimal Explanations in Probabilistic Logic Programming”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2025.
- [7] Yinxu Tang, **Stylianou Loukas Vasileiou**, and William Yeoh. “Does Your AI Agent Get You? A Personalizable Framework for Approximating Human Models from Argumentation-based Dialogue Traces”. In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2025.
- [8] **Stylianou Loukas Vasileiou**, Ashwin Kumar, William Yeoh, Tran Cao Son, and Francesca Toni. “Dialectical Reconciliation via Structured Argumentative Dialogues”. In: *International Conference on Principles of Knowledge Representation and Reasoning (KR)*. 2024.
- [9] **Stylianou Loukas Vasileiou**, Xu Borong, and William Yeoh. “A Logic-based Framework for Explainable Agent Scheduling Problems”. In: *European Conference on Artificial Intelligence (ECAI)*. 2023.
- [10] **Stylianou Loukas Vasileiou** and William Yeoh. “PLEASE: Generating Personalized Explanations in Human-Aware Planning”. In: *European Conference on Artificial Intelligence (ECAI)*. 2023.
- [11] Ashwin Kumar, **Stylianou Loukas Vasileiou**, Melanie Bancilhon, Alvitta Ottley, and William Yeoh. “VizXP: A Visualization Framework for Conveying Explanations to Users in Model Reconciliation Problems”. In: *International Conference on Automated Planning and Scheduling (ICAPS)*. 2022.
- [12] **Stylianou Loukas Vasileiou**, Alessandro Previti, and William Yeoh. “On Exploiting Hitting Sets For Model Reconciliation”. In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2021.
- [13] Son Tran Cao, Van Nguyen, **Stylianou Loukas Vasileiou**, and William Yeoh. “Model Reconciliation in Logic Programs”. In: *European Conference on Logics in AI (JELIA)*. 2021.
- [14] Van Nguyen, **Stylianou Loukas Vasileiou**, Son Tran Cao, and William Yeoh. “Conditional Updates of Answer Set Programming and Its Application in Explainable Planning (Extended Abstract)”. In: *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2020.
- [15] Van Nguyen, **Stylianou Loukas Vasileiou**, Son Tran Cao, and William Yeoh. “Explainable Planning Using Answer Set Programming”. In: *International Conference on Principles of Knowledge Representation and Reasoning (KR)*. 2020.
- [16] Son Tran Cao, Van Nguyen, **Stylianou Loukas Vasileiou**, and William Yeoh. “Conditional Updates of Logic Programs and its Applications (Poster Presentation)”. In: *International Conference on Principles of Knowledge Representation and Reasoning (KR)*. 2020.

Workshop & Symposium Papers

- [1] **Stylianou Loukas Vasileiou**, Antonio Rago, Maria Vanina Martineze, and William Yeoh. “When Humans Revise Their Beliefs, Explanations Matter: Evidence from User Studies and What It Means for AI Alignment”. In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Human-Aware and Explainable Planning*. 2025.
- [2] Yinxu Tang, **Stylianou Loukas Vasileiou**, and William Yeoh. “Approximating Human Mental Models During Argumentation-based Dialogues”. In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Human-aware and Explainable Planning*. 2024.
- [3] Silvia Tulli, **Stylianou Loukas Vasileiou**, and Sarath Sreedharan. “Human-Modeling in Sequential Decision-Making: An Analysis through the Lens of Human-Aware AI”. In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Human-aware and Explainable Planning*. 2024.
- [4] Andrew Estornell, **Stylianou Loukas Vasileiou**, William Yeoh, and Daniel Borrajo. “Predicting Customer Goals in Financial Institution Services: A Data-Driven LSTM Approach”. In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Financial Planning*. 2023.
- [5] **Stylianou Loukas Vasileiou**. “Towards a Logical Account for Human-Aware Explanation Generation in Model Reconciliation Problems”. In: *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) Doctoral Consortium*. 2023.
- [6] **Stylianou Loukas Vasileiou**, Ashwin Kumar, William Yeoh, Son Tran Cao, and Francesca Toni. “DR-HAI: Argumentation-based Dialectical Reconciliation in Human-AI Interactions”. In: *International Joint Conference on Artificial Intelligence (IJCAI) Workshop on Explainable AI*. 2023.
- [7] **Stylianou Loukas Vasileiou**, Ashwin Kumar, and William Yeoh. “On Generating Personalized Explanations via Knowledge Forgetting”. In: *International Joint Conference on Artificial Intelligence (IJCAI) Workshop on Explainable AI*. 2022.

- [8] **Stylianou Loukas Vasileiou**, William Yeoh, Son Tran Cao, and Alessandro Previti. “Explanations as Model Reconciliation via Probabilistic Logical Reasoning”. In: *International Conference on Principles of Knowledge Representation and Reasoning (KR) Workshop on Explainable Logic*. 2021.
- [9] **Stylianou Loukas Vasileiou**, William Yeoh, and Tran Cao Son. “On the Relationship Between KR Approaches for Explainable Planning”. In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Explainable AI Planning*. 2020.
- [10] **Stylianou Loukas Vasileiou**, Son Tran Cao, and William Yeoh. “A Preliminary Logic-based Approach for Explanation Generation”. In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Explainable AI Planning*. 2019.

Technical Reports

- [1] Hendrik Baier, Mark T Keane, Sarath Sreedharan, Silvia Tulli, Abhinav Verma, and **Stylianou Loukas Vasileiou**. “Explainable AI for Sequential Decision Making (Dagstuhl Seminar 24372)”. In: *Dagstuhl Reports* 14.9 (2025), pp. 67–103.

INVITED TALKS

- ICAPS Launchpad Workshop 2025

ADVISING

Ph.D. Students

- Shahriar Rahman Dipon SP’26 – NOW
New Mexico State University
- Subomi Oyewole (co-advised with Juie Shetye, Astronomy) FL’25 – NOW
New Mexico State University

M.S. Students

- Yuxiao Wang SP’23 – FL’23
Washington University in St. Louis
- Ruoyao Yang SP’20 – FL’20
Washington University in St. Louis

B.S. Students

- Benjamin Watkins, SP’23 - SU’24
Washington University in St. Louis
- Bob Xiao FL’23 – SP’23
Washington University in St. Louis
- Tanish Agarwal, SU’22
Washington University in St. Louis (REU)
- Borong Xu FL’22 – SP’23
Washington University in St. Louis

TEACHING

New Mexico State University

- CS-5405: Artificial Intelligence I SP’26
- CS-5996: Explainable AI Decision-Making FL’25

SELECTED SERVICE

Workshop Organizing Committee

- IJCAI Workshop on Explainable AI (XAI) 2025 – NOW
- ICAPS Workshop on Human-Aware and Explainable Planning (HAXP) 2021 – NOW

Program Committee

- International Joint Conference on Artificial Intelligence (IJCAI) 2023 – NOW
- Association for the Advancement of Artificial Intelligence (AAAI) 2024 – NOW
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2020, 2023 – NOW
- International Conference on Automated Planning and Scheduling (ICAPS) 2022, 2024 – NOW
- International Conference on Principles of Knowledge Representation and Reasoning (KR) 2026 – NOW
- International Conference on Learning Representations (ICLR) 2025
- European Conference on Artificial Intelligence (ECAI) 2023, 2025
- Artificial Intelligence Journal (AIJ) 2023 – NOW
- Neural Computing and Applications (NCAA) 2025 – NOW

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